

09 Orthogonality



9.1 2×2 Orthogonal Matrices

From the identity matrix to a rotation matrix — a planar geometric view of orthogonal matrices, orthogonal projection, and orthogonal complements

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

The identity, rotation, reflection, and permutation matrices — and their combinations — are all orthogonal matrices

3

Use orthogonal projection to obtain a vector's projection onto a direction

5

The orthogonal complement of a subspace contains every vector perpendicular to it

7

The Gram matrix is the core tool for judging an orthonormal basis or an orthogonal matrix

2

The identity matrix represents the identity transform on the standard orthonormal basis

4

In the plane, the sum of orthogonal projections can losslessly restore the original vector

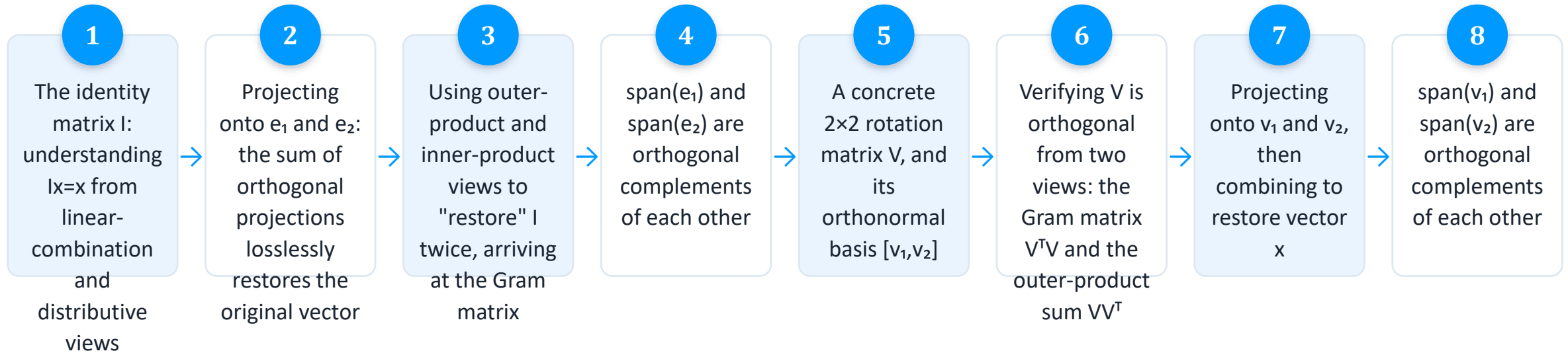
6

Understand the same matrix multiplication from four views: linear combination, distributivity, outer product, inner product

8

The columns $[v_1, v_2]$ of a 2×2 rotation matrix V also form an orthonormal basis

From the Identity Matrix to a Rotation Matrix — Understanding 2×2 Orthogonal Matrices



Seven Lenses for Every Topic in This Chapter

1 Geometric view: the identity, rotation, reflection, permutation matrices, and their combinations are all orthogonal

2 Algebraic property: $\mathbf{Q}^T = \mathbf{Q}^{-1}$ — transpose equals inverse; the determinant is always +1 or -1

3 Gram-matrix view: the columns are pairwise orthogonal, and each column is a unit vector

4 Projection view: using an orthogonal matrix's columns to project — the sum of projections restores the vector

5 Orthogonal-complement view: every subspace has an orthogonal complement perpendicular to it everywhere

6 How Gram–Schmidt orthogonalization and QR decomposition build an orthonormal basis from any basis

7 Where orthogonal matrices show up: regression, spectral decomposition, SVD, principal component analysis

Multiplying the identity matrix I by a column vector x returns x

$$Ix = [1, 0; 0, 1] @ [x_1; x_2] = [x_1; x_2] = x$$

- This "trivial" matrix multiplication hides real substance — this section revisits it through four different lenses
- The identity matrix is the simplest, most intuitive orthogonal matrix

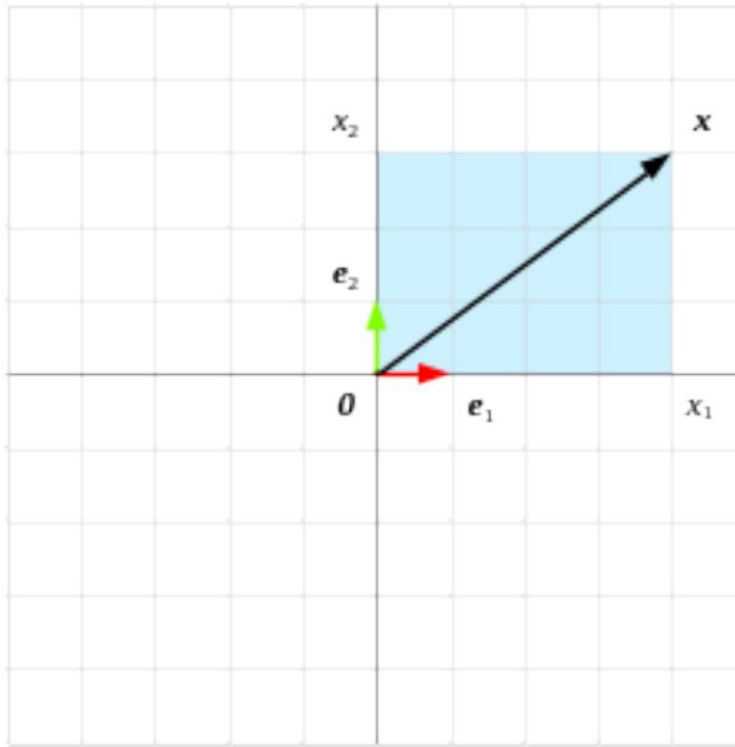
From the linear-combination view, $I\mathbf{x}$ can be written as

$$I\mathbf{x} = x_1[1;0] + x_2[0;1] = x_1\mathbf{e}_1 + x_2\mathbf{e}_2$$

- (x_1, x_2) is the coordinate of vector $\mathbf{x}$ under the standard orthonormal basis $[\mathbf{e}_1, \mathbf{e}_2]$

Figure 1 · The Plane Spanned by $[e_1, e_2]$

- e_1 and e_2 are pairwise orthogonal unit vectors; (x_1, x_2) is x 's coordinate under this basis



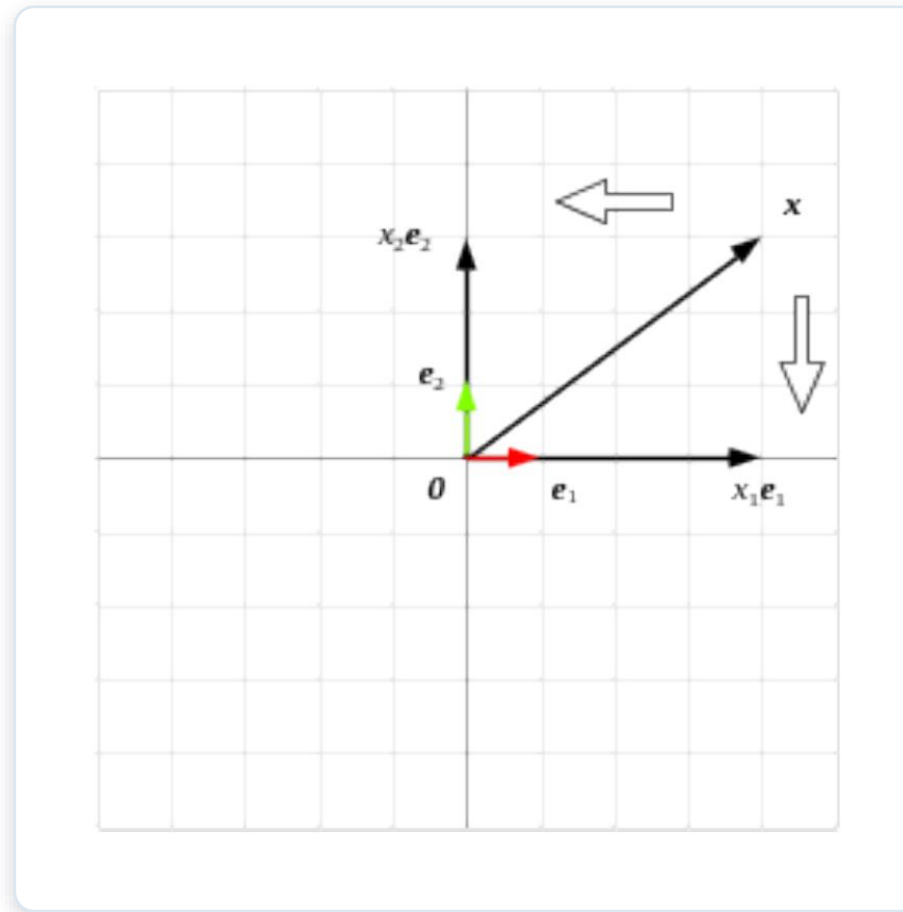
Matrix multiplication is distributive when shapes match, so I splits into a sum of two matrices

$$I\mathbf{x} = ([1,0;0,0] + [0,0;0,1])\mathbf{x} = [1,0;0,0]\mathbf{x} + [0,0;0,1]\mathbf{x} = [x_1;0] + [0;x_2]$$

- The two resulting terms are exactly $\mathbf{x}$'s orthogonal projections under the standard basis $[\mathbf{e}_1, \mathbf{e}_2]$

Figure 2 · Orthogonal Projection of $\mathbf{x}$ onto the Standard Basis $[\mathbf{e}_1, \mathbf{e}_2]$

- $\mathbf{x}$ decomposes into the sum of its projection onto $\mathbf{e}_1$ and its projection onto $\mathbf{e}_2$



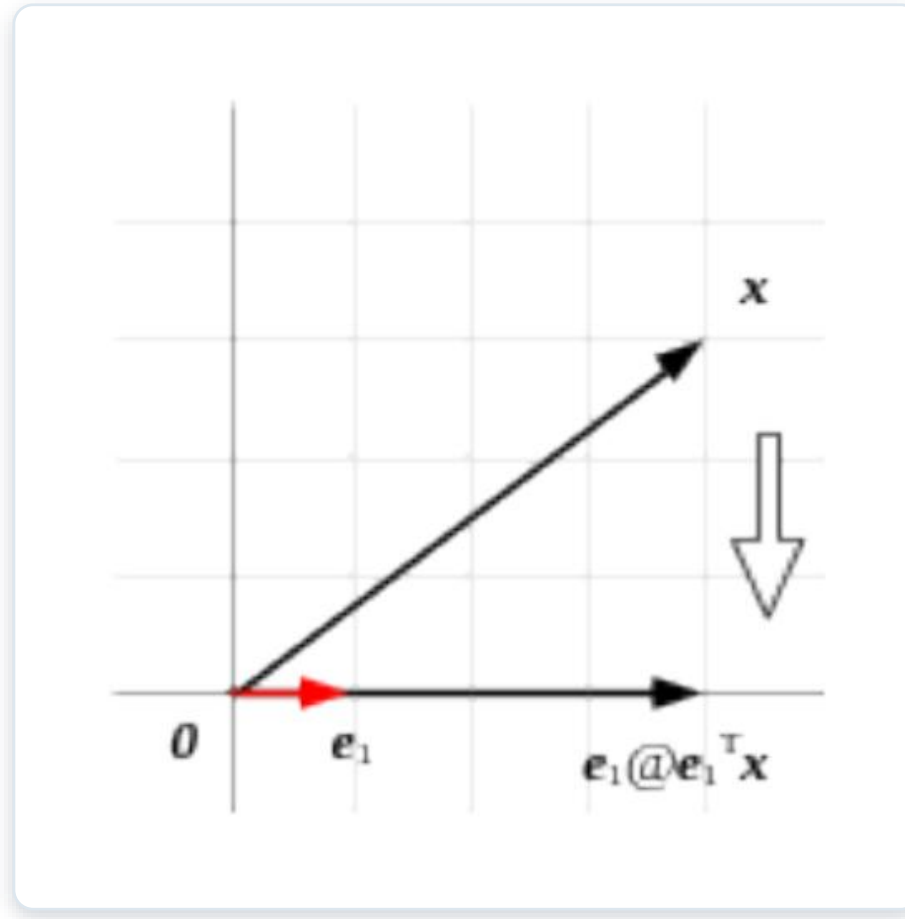
Projecting x onto the unit vector e_1 , the projection matrix P_1 is

$$P_1 = e_1 @ e_1^T = [1; 0] @ [1; 0]^T = [1, 0; 0, 0]$$

- This is exactly the projection-matrix construction introduced in §8.5 Orthogonal Projection — built from the outer product of a unit vector

Figure 3 · Orthogonal Projection of $\mathbf{x}$ onto $\mathbf{e}_1$

- The result $\mathbf{P}_1\mathbf{x} = [x_1; 0]$ keeps only the component of $\mathbf{x}$ along $\mathbf{e}_1$



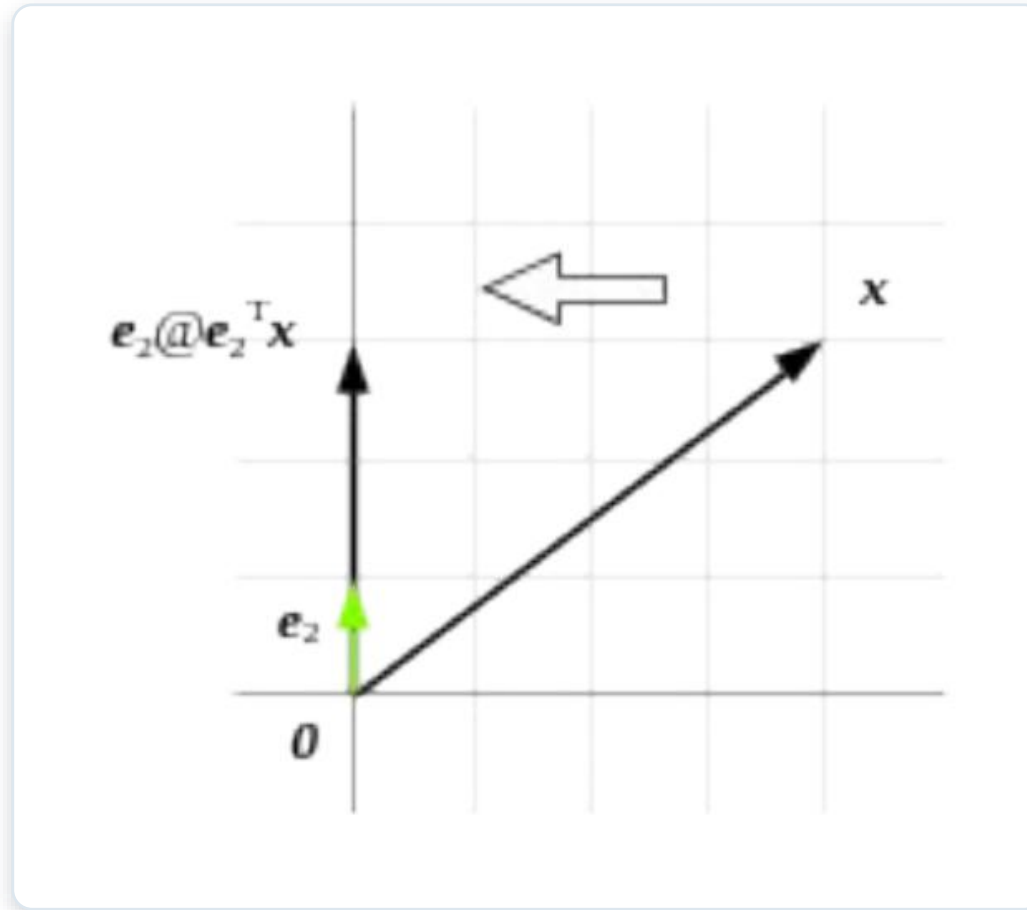
Projecting x onto the unit vector e_2 , the projection matrix P_2 is

$$P_2 = e_2 @ e_2^T = [0; 1] @ [0; 1]^T = [0, 0; 0, 1]$$

- Both P_1 and P_2 are rank-one matrices, and $P_1 + P_2$ equals I

Figure 4 · Orthogonal Projection of x onto e_2

- The result $P_2x = [0; x_2]$ keeps only the component of x along e_2



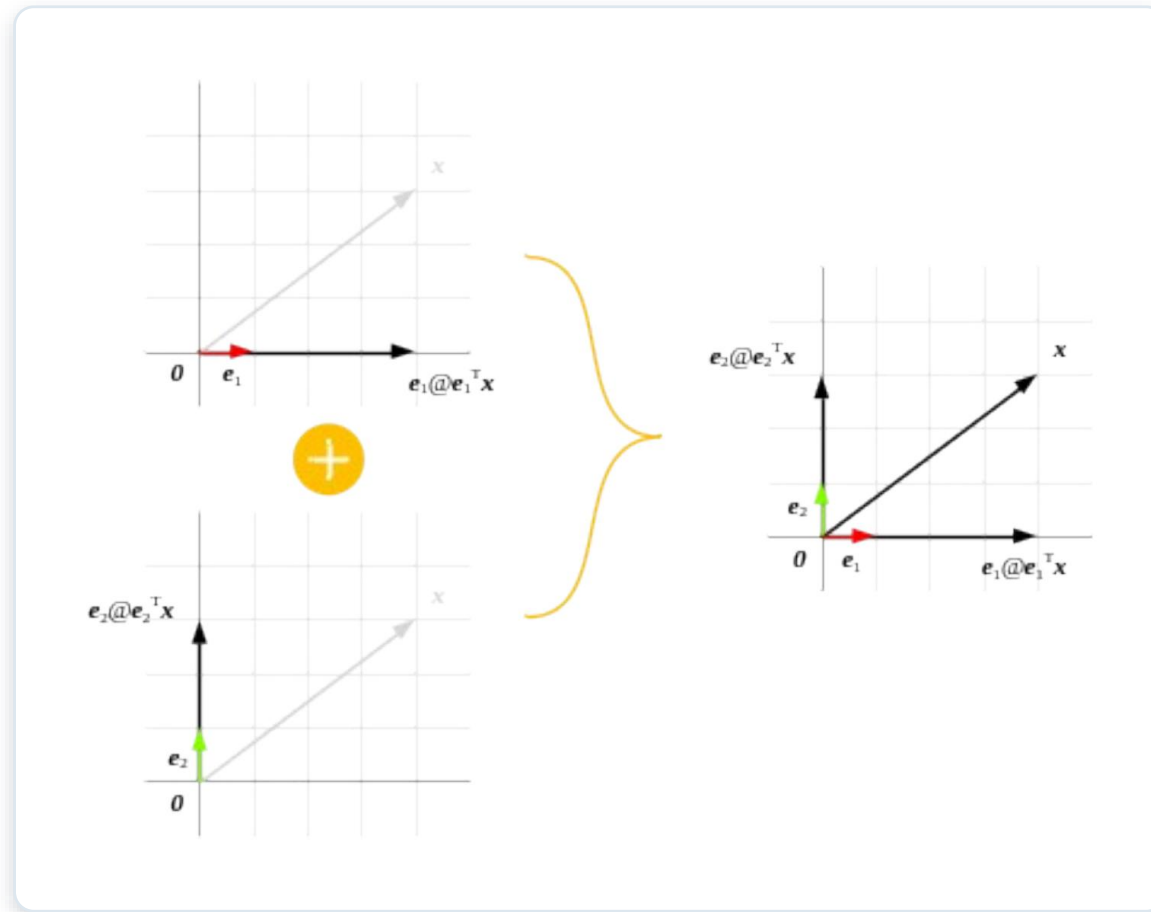
P_1 and P_2 each project down to a lower dimension; combined, they piece the original information back together

$$P_1\mathbf{x} + P_2\mathbf{x} = (P_1 + P_2)\mathbf{x} = (\mathbf{e}_1 @ \mathbf{e}_1^T + \mathbf{e}_2 @ \mathbf{e}_2^T)\mathbf{x} = I\mathbf{x}$$

- Under $[\mathbf{e}_1, \mathbf{e}_2]$, $\mathbf{x}$ is decomposed orthogonally; run it backward, and vector addition composes $\mathbf{x}$ back together

Figure 5 · Restoring the 2D Column Vector x

- Adding the projection onto e_1 to the projection onto e_2 losslessly restores the original vector x



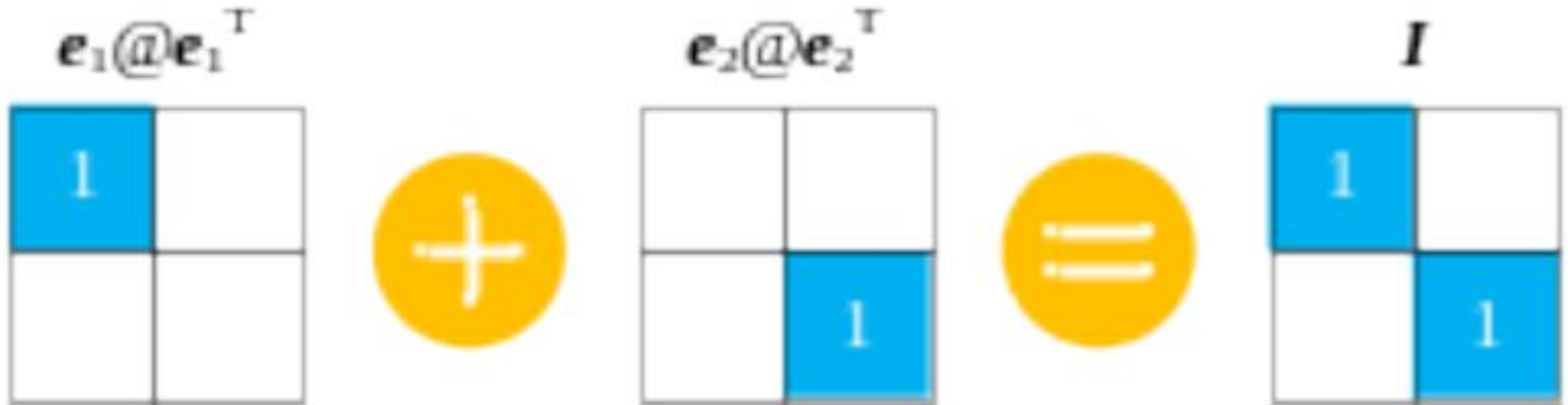
Writing the unit vectors as a row $[e_1 \ e_2]$, consider this "odd-looking" product

$$I @ I^T = [e_1 \ e_2] @ [e_1^T; e_2^T] = e_1 @ e_1^T + e_2 @ e_2^T = I$$

- This uses matrix multiplication's second view — the outer-product view

Figure 6 · Restoring the Identity Matrix

- The two rank-one matrices $\mathbf{e}_1 @ \mathbf{e}_1^T$ and $\mathbf{e}_2 @ \mathbf{e}_2^T$ add up to exactly the identity matrix I



Now look at the identity's transpose times itself, expanded via the inner-product view

$$I^T @ I = [\mathbf{e}_1^T; \mathbf{e}_2^T] @ [\mathbf{e}_1 \ \mathbf{e}_2] = [\|\mathbf{e}_1\|^2, \mathbf{e}_1 \cdot \mathbf{e}_2; \mathbf{e}_2 \cdot \mathbf{e}_1, \|\mathbf{e}_2\|^2] = I$$

- $I^T @ I$ is exactly the Gram matrix of the identity matrix I
- The diagonal holds the inner products of $\mathbf{e}_1, \mathbf{e}_2$ with themselves — the squared L^2 norms; the off-diagonal holds the pairwise inner products

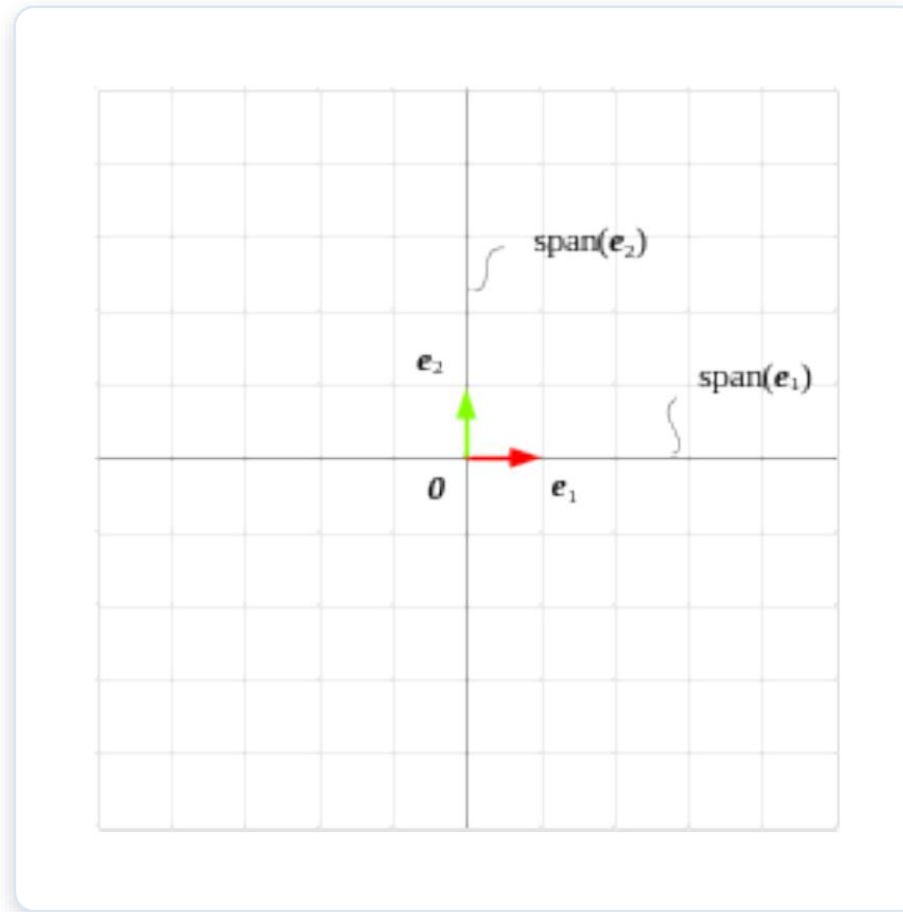
Combining the two identities above, the identity matrix is also orthogonal

$$I @ I^T = I^T @ I = I$$

- Standard basis vectors are unit length and pairwise orthogonal — together they form the most natural unit-square grid
- This can also be written $\mathbf{e}_1 @ \mathbf{e}_1^T = I - \mathbf{e}_2 @ \mathbf{e}_2^T$ — does this recall defining a projection matrix from a normal vector?
- In the plane, the line $x_2=1$ has normal $\mathbf{e}_2$, and $x_1=1$ has normal $\mathbf{e}_1$ — this will become clearer with the 3×3 identity matrix in the next section

Figure 7 · $\text{span}(e_1)$ and $\text{span}(e_2)$ Are Orthogonal Complements

- $\text{span}(e_1)$ and $\text{span}(e_2)$ are both one-dimensional; an orthogonal complement contains every vector perpendicular to the original space



A 2×2 ROTATION MATRIX

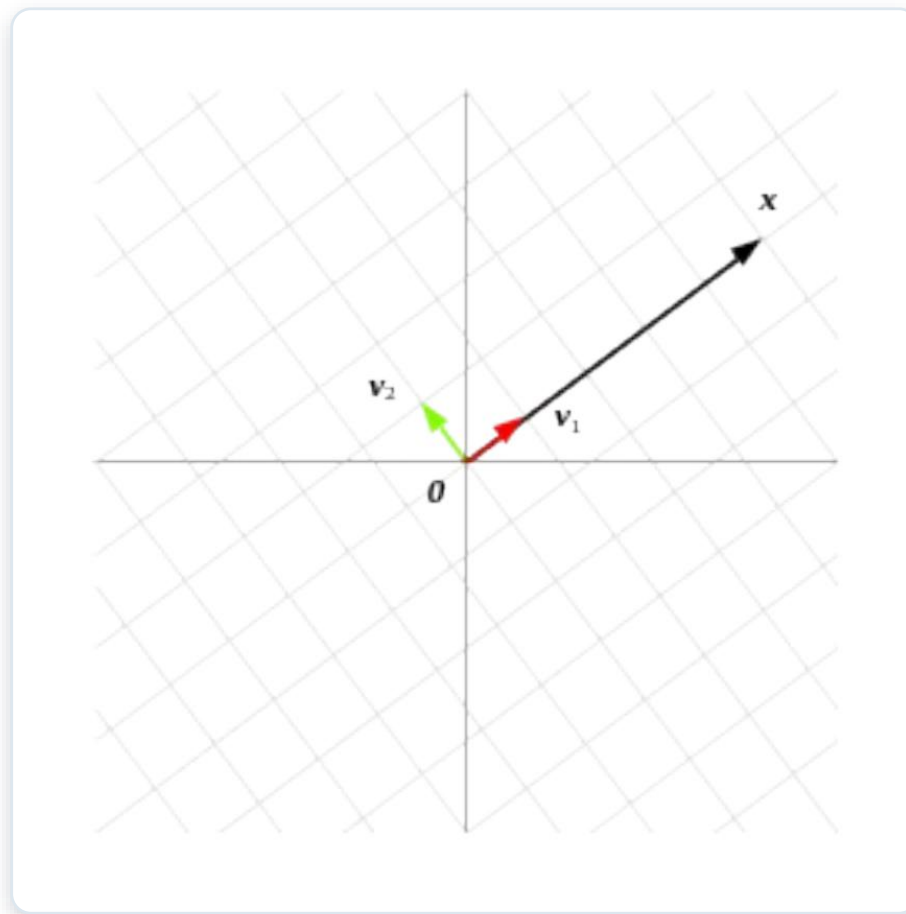
As mentioned earlier, a rotation matrix is also orthogonal — here is a concrete 2×2 rotation matrix V

$$V = [4/5, -3/5; 3/5, 4/5] = [\mathbf{v}_1 \ \mathbf{v}_2]$$

- $\mathbf{v}_1 = [4/5, 3/5]^T$ and $\mathbf{v}_2 = [-3/5, 4/5]^T$ are both unit length and mutually perpendicular
- $[\mathbf{v}_1, \mathbf{v}_2]$ is also an orthonormal basis

Figure 8 · The Plane Spanned by $[v_1, v_2]$

- Vector x also has its own coordinate under the orthonormal basis $[v_1, v_2]$



Because V is orthogonal, it satisfies

$$V@V^T = V^T@V = [1,0; 0,1] = I$$

- The Gram matrix $V^T@V$, expanded via the inner-product view: the diagonal holds $\|v_1\|^2$, $\|v_2\|^2$, the off-diagonal their inner product
- $V@V^T$, expanded via the outer-product view, is the sum of two projection matrices $v_1@v_1^T$ and $v_2@v_2^T$, which also equals I

Projecting the 2D column vector x onto the unit vector v_1 , the projection matrix P_1 is

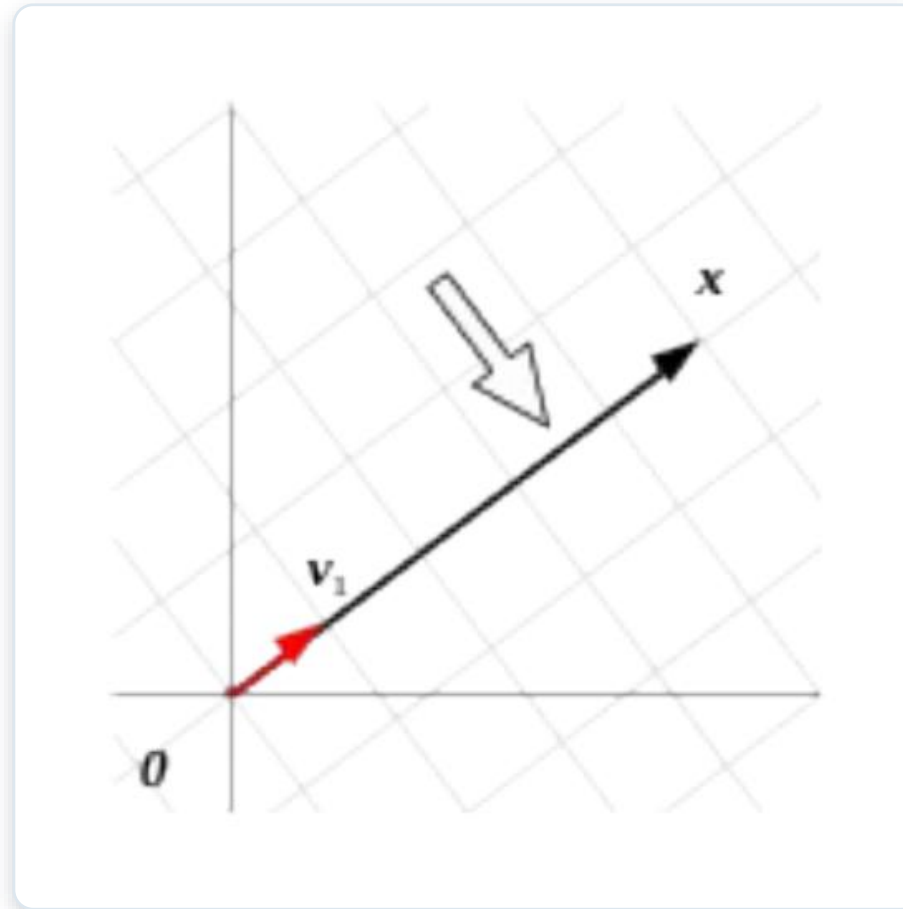
$$P_1 = v_1 @ v_1^T = [4/5; 3/5] @ [4/5; 3/5]^T = [16/25, 12/25; 12/25, 9/25]$$

- Substituting $x=[4;3]$, the projection $P_1 x=[4;3]$ — x itself

X IS PARALLEL TO v_1

Figure 9 · Orthogonal Projection of x onto v_1

- x is parallel to v_1 — that is, x lies in $\text{span}(v_1)$ — so the projection is x itself
- The scalar projection $v_1^T x = 5$ is x 's coordinate along v_1 ; multiplying by v_1 "re-projects" it onto the plane



Projecting the 2D column vector x onto the unit vector v_2 , the projection matrix P_2 is

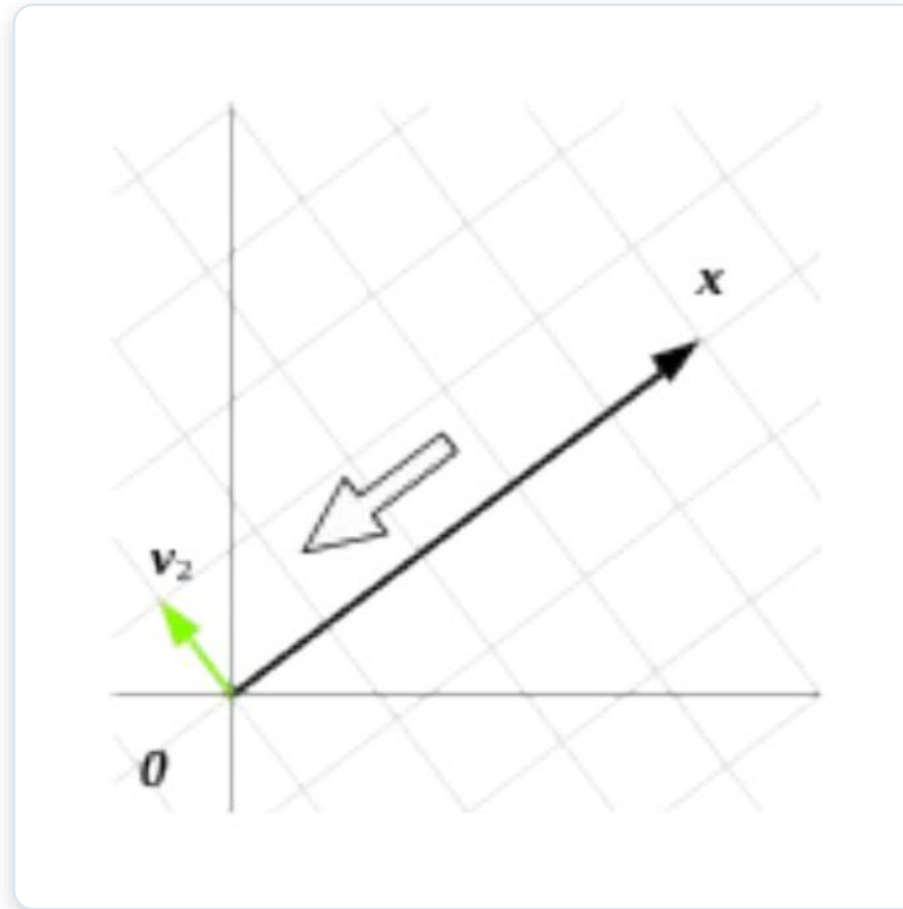
$$P_2 = v_2 @ v_2^T = [-3/5; 4/5] @ [-3/5; 4/5]^T = [9/25, -12/25; -12/25, 16/25]$$

- Substituting $x=[4;3]$, the projection $P_2 x=[0;0]$ — the zero vector

X IS ORTHOGONAL TO v_2

Figure 10 · Orthogonal Projection of x onto v_2

- Since x and v_2 are orthogonal, x 's projection onto v_2 is the zero vector 0



Adding the projections onto v_1 and v_2 restores all of x 's information

$$(P_1 + P_2)x = (v_1 @ v_1^T + v_2 @ v_2^T)x = Ix = x$$

- Even for an orthonormal basis that isn't the standard one, the sum of orthogonal projections still losslessly restores the original vector

Figure 11 · Restoring the 2D Column Vector x Under $[v_1, v_2]$

- Adding the projection onto v_1 to the projection onto v_2 losslessly restores the original vector x

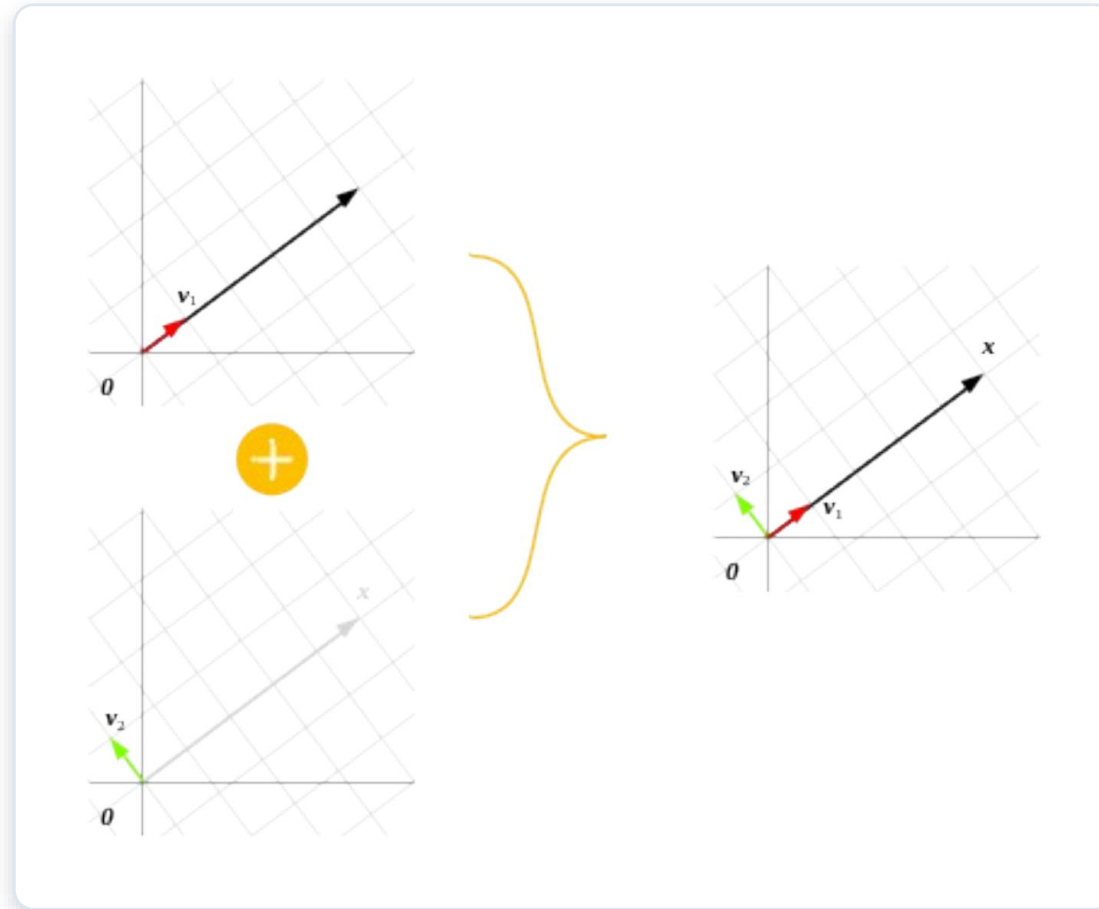
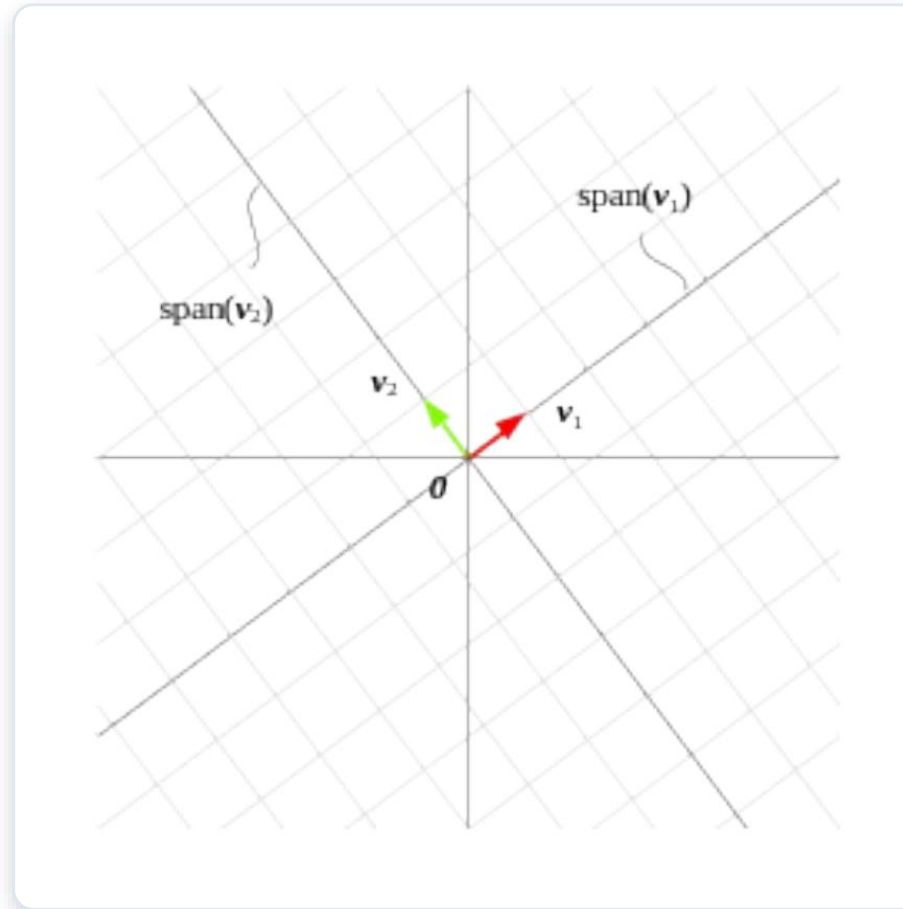


Figure 12 · $\text{span}(v_1)$ and $\text{span}(v_2)$ Are Orthogonal Complements

- Whether it's the standard basis or a rotated orthonormal basis, the two one-dimensional spans are always orthogonal complements



SUMMARY

What Do the Four Views of Matrix Multiplication Reveal About Orthogonal Matrices?

The identity matrix is the simplest orthogonal matrix: its columns $[\mathbf{e}_1, \mathbf{e}_2]$ are the standard orthonormal basis

Linear combination, distributivity, outer product, inner product — four views of the same underlying orthogonal relationship

The Gram matrix $\mathbf{V}^T \mathbf{V} = \mathbf{I}$ is the core tool for judging an orthonormal basis or an orthogonal matrix

The sum of orthogonal projections restores the vector: $\mathbf{P}_1 + \mathbf{P}_2 = \mathbf{I}$, matching $\text{span}(\mathbf{v}_1)$ and $\text{span}(\mathbf{v}_2)$ as orthogonal complements

Glossary

Orthogonal Matrix 正交矩阵 —— a square matrix satisfying $Q^T @ Q = Q @ Q^T = I$

Orthonormal Basis 规范正交基 —— a set of vectors that are pairwise orthogonal and unit length

Orthogonal Projection 正交投影 —— $P = u @ u^T$, the projection of a vector onto unit vector u

Rotation Matrix 旋转矩阵 —— an orthogonal matrix with determinant +1 that preserves length and angle

Inner Product 内积 —— the vector dot product, matrix multiplication's first view, measuring angle and projection

Standard Orthonormal Basis 标准正交基 —— $[e_1, e_2, \dots]$, the unit vectors along the coordinate axes

Gram Matrix 格拉姆矩阵 —— $V^T @ V$, diagonal holds squared lengths, off-diagonal holds pairwise inner products

Orthogonal Complement 正交补 —— the space of all vectors perpendicular to every vector in a given subspace

Outer Product 外积 —— a column vector times a row vector, matrix multiplication's second view

span 张成空间 —— the space spanned by the linear combinations of a set of vectors

EXPLORE FURTHER

Further Reading

Recap · 8.5 Orthogonal Projection

Previous section — building projection matrices from normal and tangent vectors

Further Reading · 9.2 3×3 Orthogonal Matrices

Later in this chapter — extending orthogonal matrices to 3D

Further Reading · 9.4 Gram–Schmidt Orthogonalization

Later in this chapter — building orthogonal bases using projection ideas

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Matrices A – I (used in Q1 – Q6)

Label	Matrix	Label	Matrix
A	$[1,0; 0,1]$	F	$[1,1; -1,1]$
B	$[1,1; 0,1]$	G	$[0,1; 1,0]$
C	$[1,0; 0,-1]$	H	$[\sqrt{2}/2,\sqrt{2}/2; \sqrt{2}/2,\sqrt{2}/2]$
D	$[-1,0; 0,1]$	I	$[\sqrt{2}/2,-\sqrt{2}/2; \sqrt{2}/2,\sqrt{2}/2]$
E	$[-1,0; 0,-1]$		

In Q4, matrix B is always matrix I (a 45° rotation); use this section's ideas to judge which matrices are orthogonal.

Q1 – Q9

Q1 Determine which of the matrices $A - I$ in the table above are orthogonal.

Q3 Compute the determinant of each orthogonal matrix in Q1.

Q5 Geometrically, what transforms do $A@I$ and $I@A$ in Q4 correspond to?

Q7 Compute the scalar and vector projection of $[2,3]^T$ onto e_1 in the plane.

Q9 Project $[2,3]^T$ onto e_1 and e_2 separately, then combine the results and verify they restore the original vector.

Q2 Geometrically, what transform does each orthogonal matrix in Q1 correspond to?

Q4 Matrix A below and matrix I in the table above are both orthogonal; determine whether $A@I$ and $I@A$ are orthogonal: ① $A=[1,0;0,-1]$; ② $A=[0,1;1,0]$; ③ $A=[3/5,-4/5;4/5,3/5]$.

Q6 Compute the inverse and transpose of each: $[0,1;1,0]$, $[3/5,-4/5;4/5,3/5]$, $[\sqrt{3}/2,-1/2;1/2,\sqrt{3}/2]$. What pattern do you observe?

Q8 Compute the scalar and vector projection of $[2,3]^T$ onto e_2 in the plane.

Q10 – Q17

Q10 Write a Python function `is_orthogonal(M)` that checks whether a square matrix is orthogonal (verify $M @ M.T$ is approximately I).

Q12 Write code to compute the determinant of each matrix in Q1, and observe that an orthogonal matrix's determinant is always $+1$ or -1 .

Q14 Explore: what geometric transform does the permutation matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ correspond to? Verify with Python that it is orthogonal.

Q16 Verify with code that the Gram matrix $V^T V$ of the rotation matrix $V = \begin{bmatrix} 4/5 & -3/5 \\ 3/5 & 4/5 \end{bmatrix}$ equals the identity.

Q11 Use numpy to check each of the 9 matrices in Q1, and compare the results with your hand calculations.

Q13 Use matplotlib to visualize the unit square after being transformed by several orthogonal matrices from Q1.

Q15 Write a Python function `gram_matrix(V)` that computes the Gram matrix of a matrix's column vectors.

Q17 Explore: given an arbitrary unit vector $v1$, how do you construct a unit vector $v2$ orthogonal to it so that $[v1, v2]$ is a 2×2 orthogonal matrix?

Q18 – Q25

Q18 Write a Python function `orthogonal_projection(x, u)` that computes the orthogonal projection of vector x onto unit vector u .

Q20 Explore: given an orthonormal basis $[v_1, v_2]$, does the sum of x 's projections onto v_1 and v_2 also always equal x ? Verify with code.

Q22 Explore $\text{span}(e_1)$ and $\text{span}(e_2)$ as orthogonal complements: verify with code that any vector from each subspace has zero inner product.

Q24 Explore: treating the identity matrix I and rotation matrix V each as an orthonormal basis, compare the grid shapes they each define.

Q19 Verify with code: for any 2D vector x , the sum of its projections onto e_1 and e_2 always equals x .

Q21 Write code comparing vector x 's coordinates under the standard basis $[e_1, e_2]$ versus the rotated basis $[v_1, v_2]$.

Q23 Comprehensive: for any 2×2 orthogonal matrix Q , verify with Python that both columns have unit length and are mutually perpendicular.

Q25 Preview thought: the next section analyzes 3×3 orthogonal matrices in 3D — do this section's properties extend to three dimensions?

10 Prompts to Explore With AI

1 Explain why multiplying the identity matrix I by any column vector x returns x , and what views reveal this 'trivial' result.

2 Explain $Ix=x$ from both the linear-combination and distributive views of matrix multiplication.

3 Explain why $e_1 @ e_1^T + e_2 @ e_2^T = I$, and how this relates to orthogonal projection.

4 Explain what the Gram matrix $V^T V$ means, and why it can determine whether a set of vectors forms an orthonormal basis.

5 Use Python to demonstrate: projecting a 2D vector onto two orthogonal directions, then combining the results to restore the original vector.

10 Prompts to Explore With AI

6 Explain why the rotation matrix V is also orthogonal, and what geometric relationship holds between its columns.

7 Explain what it means for $\text{span}(e_1)$ and $\text{span}(e_2)$ to be orthogonal complements, and give the general definition of an orthogonal complement.

8 Explain the four views of matrix multiplication (linear combination, outer product, inner product, distributivity) and their geometric meanings.

9 Explain why an orthogonal matrix's columns must be both pairwise orthogonal and unit length — is either condition alone enough?

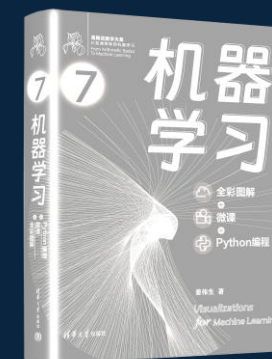
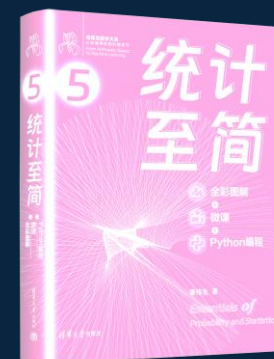
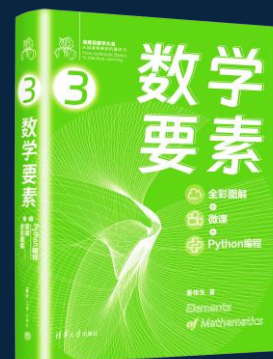
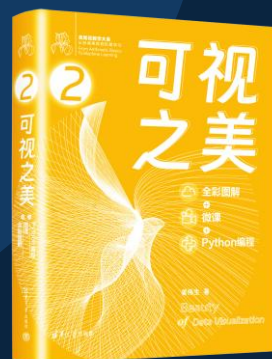
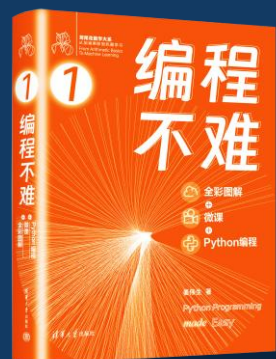
10 Explain what applications orthogonal matrices might have in Gram–Schmidt orthogonalization, QR decomposition, and regression analysis.

数学

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概率统计



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工具

数据分析
图和网络

回归、降维
分类、聚类

实践

